Combining all conditional counts by one crossing identity

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Separate upper bounds on incoming counts discard cancellation with outgoing counts. The following identity retains every count of the same iid law. Its leading loss has the correct sign for all noise levels. The proof is a finite calculation; no Poisson theorem, tail truncation, or parameter grid enters it. The unequal binomial weights still require a separate estimate.

Theorem 1. *Let U_0, U_1 be integrable, centered conditional-noise laws with*

$$U_0 \le a_0, \qquad -a_1 \le U_1 \le A_1, \quad a_0, a_1, A_1 \ge 0, \quad a_0 + a_1 < 1, \quad A_1 + a_1 \le 1.$$

Take $2n$ mutually independent variables U_{0i}, U_{1i} with these laws, and define

$$T_j = j + \sum_{i \le j} U_{1i} + \sum_{i > j} U_{0i}, \quad 0 \le j \le n.$$

If $na_0 \le t < n(1 - a_1)$, there is a unique $K \in \{0, \dots, n-1\}$ with $T_K \le t < T_{K+1}$. Put

$$b = \min\{1 - A_1 - a_1, 1 - a_0 - a_1\} \ge 0, \qquad R = t - T_K.$$

Then

$$g_n(t) := t - \mathbb{E}K \;\ge\; \mathbb{E}\min\{R, b\} \;\ge 0. \tag{1}$$

More precisely, let $D = 1 + U_1$ and define the nonnegative kernel

$$h(x) = \mathbb{E}(x - U_0)_+ - \mathbb{E}(x - D)_+ - \mathbb{P}(D \le x),$$
$$R_i = t - (i-1) - \sum_{l<i} U_{1l} - \sum_{l>i} U_{0l}.$$

The exact cancellation identity is

$$g_n(t) = \sum_{i=1}^{n} \mathbb{E}h(R_i). \tag{2}$$

Proof. Each increment $T_i - T_{i-1} = 1 + U_{1i} - U_{0i}$ is at least $1 - a_0 - a_1 > 0$. The threshold assumptions give $T_0 \le t < T_n$. Thus the closed lower and strict upper crossing events partition the sample space, even when t is an atom of a conditional sum.

Set $m = 1 - a_1$. If $x < m$, both terms involving D in the definition of h vanish, so

$$h(x) = \mathbb{E}(x - U_0)_+ \ge 0. \tag{3}$$

If $x \ge m > a_0$, then $\mathbb{E}(x - U_0)_+ = x$. Since $\mathbb{E}D = 1$, put–call parity and the bound $D \le 1 + A_1$ give

$$\begin{aligned} h(x) &= \mathbb{P}(D > x) - \mathbb{E}(D - x)_+ \\ &\ge (1 - A_1 - a_1)\mathbb{P}(D > x) \ge b\mathbb{P}(D > x). \end{aligned} \tag{4}$$

This includes $x = m$ with its strict survival probability.

Let $L_j(t) = \mathbb{E}(t - T_j)_+$. The variables omitted from R_i are independent of it. Consequently the definition of h gives

$$\mathbb{E}h(R_i) = L_{i-1}(t) - L_i(t) - \mathbb{P}(T_i \le t). \tag{5}$$

Sum over i. The stop-loss expectations telescope, with $L_0(t) = t - \mathbb{E}T_0 = t$ and $L_n(t) = 0$. Moreover $\mathbb{E}K = \sum_{i=1}^n \mathbb{P}(T_i \le t)$. This proves (2) and $g_n(t) \ge 0$.

For the stronger age bound, condition on $R_i = x$. Crossing at i means $U_{0i} \le x < D_i$, and its age is $x - U_{0i}$. If $x < m$, then $D_i > x$ surely, and (3) is at least the conditional expected value of $\min\{x - U_{0i}, b\}$ on this event. If $x \ge m$, then $U_{0i} \le a_0 < x$ surely, the crossing probability is $\mathbb{P}(D_i > x)$, and its age is at least $m - a_0 \ge b$. Now (4) applies. Summing over the unique crossing proves (1). All expectations are finite, and no stopping-time or limiting argument has been used. □

An exact replacement for the separate count sums

For $Y = B + U$ with $\mathbb{P}(B = 1) = p$ and these conditional laws, the conditional distribution of a raw iid sum given j successes is precisely that of T_j. With F_B the $\mathrm{Bin}(n, p)$ CDF,

$$\mathbb{P}(Y_1 + \cdots + Y_n \le t) = \mathbb{E}F_B(K). \tag{6}$$

The coupling is auxiliary: it preserves each conditional law, hence preserves the actual iid mixture. Only those conditional marginals are needed; no claim about the original joint law of its count and K is made.

For the acquired right strip, $a_0 + a_1 \le .072$ and $A_1 = .6$, one may take $b = .328$ in (1); for the left strip, inward sum at most .068 and $A_1 = .7$ give $b = .232$. For $n \ge 42$, $p \in [.25, .53]$, $s = \mathbb{E}U^2$, $ns \le .616$ and $|H| \le 1.84$, put $t = np + H\sqrt{p(1-p) + s}$. Then

$$.25n - .948 < t < .53n + .948, \qquad na_0 \le t < n(1 - a_1).$$

Indeed $\sqrt{1/4 + .616/42} < .515$ and $1.84(.515) < .948$; the remaining two inequalities hold already at $n = 42$. Thus the cancellation identity covers the whole retained acquired noise range. It makes no assertion that the acquisition hypotheses themselves hold without their existing contact and predecessor dependencies.

The weighted identity isolates the remaining difficulty

Write $\pi_j = \mathbb{P}(B_n = j)$, $L_j(t) = \mathbb{E}(t - T_j)_+$, $B_j(t) = L_j(t) - (t - j)_+ \ge 0$ (by Jensen), and $g_i(t) = \mathbb{E}h(R_i) \ge 0$. Multiply (5) by π_i and sum by parts. Because the count-zero CDF is one and $L_0 = t$, $L_n = 0$, this gives

$$\mathbb{P}(Y_1 + \cdots + Y_n \le t) = \pi_0 + \pi_1 t + \sum_{j=1}^{n-1} (\pi_{j+1} - \pi_j) L_j(t) - \sum_{i=1}^{n} \pi_i g_i(t). \tag{7}$$

Let $k = \lfloor t \rfloor$ and $u = t - k \in [0, 1)$. Replacing L_j by $(t - j)_+$ in the first three terms gives the linear interpolation $F_B(k) + u\pi_{k+1}$. Hence the exact formula is

$$\mathbb{P}(Y_1 + \cdots + Y_n \le t) - F_B(k) = u\pi_{k+1} + \sum_{j=1}^{n-1} (\pi_{j+1} - \pi_j) B_j(t) - \sum_{i=1}^{n} \pi_i g_i(t). \tag{8}$$

Both B_j and g_i are nonnegative. Only the increasing side of the binomial mass function contributes unfavorably to the curvature sum. No separate incoming-count tail appears.

For the following normalized comparison assume additionally $0 < p < 1$ and $\mathbb{E}|U_0|^3 + \mathbb{E}|U_1|^3 < \infty$, as holds in the acquired bounded strips. Put $v = p(1-p)$, $s = \mathbb{E}U^2$, $\rho = \mathbb{E}|Y-p|^3/(v+s)^{3/2}$ and $\rho_B = (1-2v)/\sqrt{v}$. For the direct comparison at this integer anchor,

$$\begin{aligned}\Delta_C = & -\sqrt{n}\sum_{i=1}^{n}\pi_i g_i(t) + \sqrt{n}\sum_{j=1}^{n-1}(\pi_{j+1}-\pi_j)B_j(t) + \sqrt{n}\,u\pi_{k+1} \\ & -\sqrt{n}\left[\Phi\left(\frac{t-np}{\sqrt{n(v+s)}}\right) - \Phi\left(\frac{k-np}{\sqrt{nv}}\right)\right] + C(\rho_B - \rho). \end{aligned} \tag{9}$$

Every identity is exact at the actual sample size. The remaining task is to absorb the binomial curvature and the phase, normal and cubic repairs in the common nonnegative loss. A sign for the unweighted loss alone does not do this. Therefore the current general count certificates remain ancestors of the accepted comparison until that joint inequality is proved.

For comparison with a direct stopped-sum calculation, put $W_j = T_j - j$. Its selected mean is generally nonzero:

$$\mathbb{E}W_{K+1} = -\sum_{i=1}^{n}\mathbb{E}\big[U_{0i}\mathbf{1}_{\{T_{i-1}\le t\}}\big] \ge 0.$$

The sign follows by conditioning on the omitted-pair sum and centering U_{0i}. The telescoping proof includes this bias automatically; setting it to zero would be an unjustified use of optional stopping.

Controls and limitations

The companion exact checks reconstruct all conditional sums and the local kernel h separately, including closed collisions and strict crossings. They check (2), (1), (6), (8) and the nonzero selected-noise bias. They also include a false extension: take $U_0 = 0$ and $D = 9/10, 39/20$ with probabilities $19/21, 2/21$. At $n = 2$, $t = 9/10$, $\mathbb{E}K - t = 1/210 > 0$; here $A_1 + a_1 = 21/20 > 1$. No assertion of strict gain proportional to s, or of the full sharp Berry–Esseen bound, follows from this note alone.